Rigression and Correlation of Data : Statistical Tools

Lecture: VI

Introduction

By the term data we simply mean certain quantifier, which quantifies certain values or physical properties.

For example, we can collect the data $x_1, x_2, \ldots, x_n$ regarding the marks obtained by students. Here x_i is a variable which quantifies the attribute marks by assigning different values to $x_1, x_2, \ldots, x_n$.

variable can be of two types. Dependent and independent variables.

Independent variables are those whose value does not depend on the values of any other variable. And

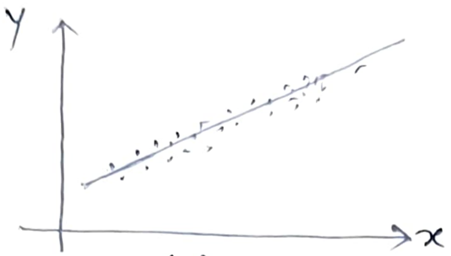
Depen

The first step is the collection of data showing correspon-ding values of the variables. For example, suppose x and y denote, respectively the height and weight of an adult male. Then a sample of n-indi-viduals would reveal the heights $x_1, x_2, \ldots, x_n$ and the corresponding weights $w_1, w_2, \ldots, w_n$.

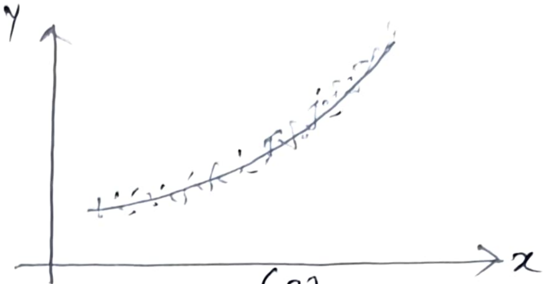
The next step is to plot the points ~~$(x_1, y_1$~~ (x_1, ω_1), $(x_2, \omega_2) \ldots, (x_n, \omega_n)$ on a rectangular coordinate-system. The resulting set of points are called a scatter diagram.

From the scatter diagram it is often possible to visua-lize a smooth curve approximating the data. Such a curve is called an approximating curve.

Ex:



(1)



(2)

(3)

In fig-1, the data appear to be approximatted well by a straight line like-

$$y = ax + b \quad \ldots \quad (1)$$

we say that linear relationship exists between the variables.

In fig-2. The relationship exists between the variables x and y is non-linear and can be filled with a curve which is parabola or quadratic curve in general.

$$y = a + bx + cx^2 \quad \ldots\ldots \quad (2)$$

In fig-3 There exists no relation between the ~~depen~~ independent and dependent variables.

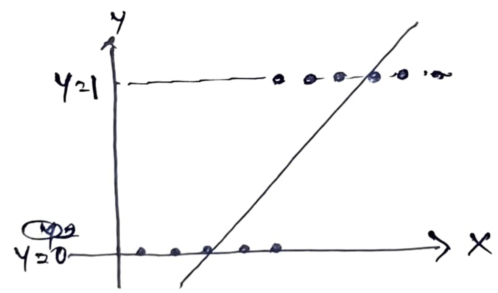
Regression: One of the main purpose of curve fitting is to estimate one of the variables (the dependent variable) from the other (the independent variable). The process of estimation is referred to as regression. If y is to be estimated from x by means of some equation, we call the equation a regression equation of y on x and the corresponding curve a regression curve of y on x.

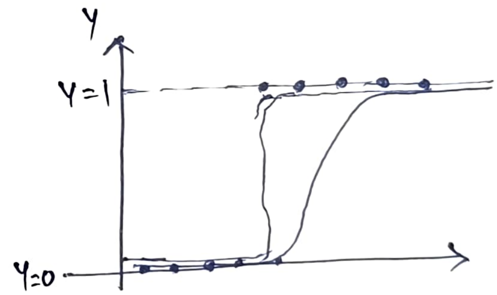
→ We normally deals with two types of regression i) Linear regression and ii) logistic regression.

Linear and logistic regression very popular in machine learning. And both of these techniques are categorized as supervised learning technique. Since both the algorithms are of supervised in nature, therefore the algorithms used labeled data set. The main difference between the linear and logistic regression is how they are being used.

The linear regression is used for prediction the values of the independent variable using the values of the dependent variable.

Logistic Regression is used for solving the classification problem.

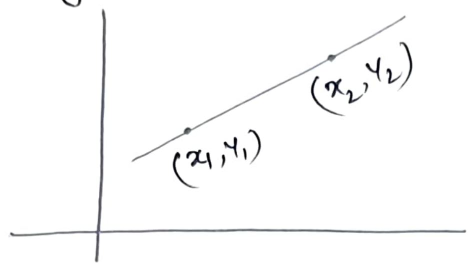




In the figures above we have represented the scattered diagram of a data set and fitted them with linear and logistic regression curve. Here we see that in this kind of data where the Y values are either 0 or 1 the linear regression model does not give the adequate results.

Now we will consider certain numerical examples to understand the concept of linear regression.

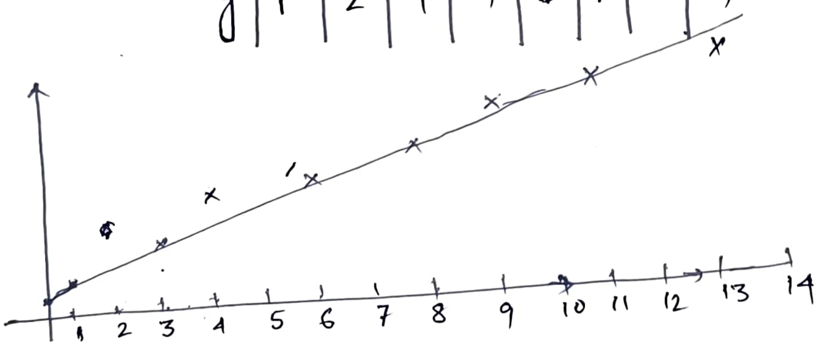
To understand the concepts of linear regression, we will assume that the equation of a straight line passing through two points (x_1, y_1) and (x_2, y_2) is —



$$\left(\frac{y - y_1}{y_2 - y_1}\right) = \left(\frac{x - x_1}{x_2 - x_1}\right) \Rightarrow y - y_1 = \frac{(y_2 - y_1)}{(x_2 - x_1)}(x - x_1)$$

Ex:-1 (a) Construct a straight line that approximates the data as given below —

x	1	3	4	6	8	9	11	14
y	1	2	4	4	5	7	8	9



To construct the equation of the line, we choose two points on the line, such as —

$P = (0, 1)$

and $Q = (12, 7.5)$

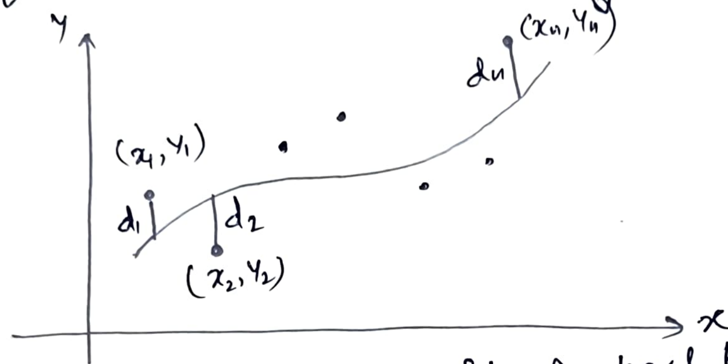
Then the equation of the straight line is —

$$y - 1 = \frac{7.5 - 1}{12 - 0}(x - 0)$$

or $y - 1 = 0.542x \Rightarrow y = 1 + 0.542x$

Least - Square Line

If we try to connect points ~~by~~ empirically by individual's own approximation, then we will get more than one curve to fit the data. To avoid individual judgment in constructing lines, it is necessary to agree on a definition of a "best-fitting" line.



To understand the concept of best fit line let us consider a ~~set~~ set of data points $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$

For a given value of x, say x_1, there will be a difference between the value y_1 and the corresponding value as determined from the curve C. We denote the difference by d_1, This is called as deviation, error or residual. It may be positive, negative or zero. Similarly, corresponding to the values $x_2, \ldots, x_n$ we obtain the deviations $d_2, \ldots, d_n$.

A measure of the goodness of fit of the curve C to the set of data is provided by the quantity

$$e = d_1^2 + d_2^2 + \cdots + d_n^2.$$

If e is small the fit is good and if e is large the fit is bad. We therefore make the following definition.

Definition: Of all curves in a given family of curves approximating a set of n data points, a curve having the property that

$$d_1^2 + d_2^2 + \cdots + d_n^2 = \sum_{i=1}^{n} d_i^2 = \text{minimum}.$$

is called a best-fitting curve in the family.

A curve having this property is set to fit the data in the least-squares sense is called a least-squares regression curve or simply a least-squares curve.

using the above definition we first assume that the least-squares line approximating the set of points $(x_1, y_1) \ldots, (x_n, y_n)$ has the equation

$$y = a + bx$$

To determine the constants a ~~ands~~ and b we construct the following two equations -

$$\Sigma y = an + b\Sigma x \quad -(1)$$

and $$\Sigma xy = a\Sigma x + b\Sigma x^2 \quad -(2)$$

These two equations are called the normal equations for the best-squares line.

solving eqn. (1) and (2) we get -

$$\Sigma x^2 \Sigma y = an\Sigma x^2 + b\Sigma x\Sigma x^2 \quad -(3)$$

$$\Sigma x\Sigma xy = a(\Sigma x)^2 + b\Sigma x\Sigma x^2 \quad -(4)$$

subtracting (4) from (3)

$$\therefore \Sigma y\Sigma x^2 - \Sigma x\Sigma xy = a[n\Sigma x^2 - (\Sigma x)^2]$$

$$\Rightarrow a = \frac{(\Sigma y)(\Sigma x^2) - (\Sigma x)(\Sigma xy)}{n\Sigma x^2 - (\Sigma x)^2}$$

Again $$\Sigma x\Sigma y = an\Sigma x + b(\Sigma x)^2$$

$$n\Sigma xy = an(\Sigma x) + bn(\Sigma x^2)$$

$$n\Sigma xy - \Sigma x\Sigma y = b[n\Sigma x^2 - (\Sigma x)^2]$$

$$\Rightarrow b = \frac{n\Sigma xy - (\Sigma x).(\Sigma y)}{n\Sigma x^2 - (\Sigma x)^2}$$

Let us now consider a numerical example -

The data set is -

x	1	3	4	6	8	9	11	14
y	1	2	4	4	5	7	8	9

$\sum x = 56 \quad \sum y = 40.$

$\therefore$ using the normal Eqn

$$\sum y = an + b\sum x$$

$\Rightarrow \quad 8a + 56b = 40 \qquad \because$ there are 8 set of data $n = 8$.

and $\sum xy = a\sum x + b\sum x^2$

$\Rightarrow \cancel{56}\ 364 = 56a + 524b.$

solving - $\quad 8a + 56b = 40$

$\quad \& \ 56a + 524b = 364$

gives $a = \frac{6}{11}$ or 0.545 and $b = \frac{7}{11}$ or 0.636.

and the required list-square line is

$y = 0.545 + 0.636x.$

$y = \frac{6}{11} + \frac{7}{11}x$ or $y = 0.545 + 0.636x$

1 —

Linear Regression
Problem set

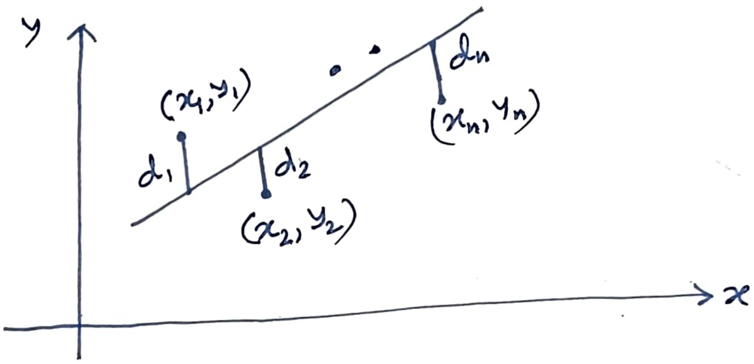
Ex: 1 The least-squares line approximating the set of points $(x_1, y_1), \ldots, (x_n, y_n)$ has the equation

$$y = a + bx$$

The constant a and b are determined by solving ~~the~~ simultaneously the normal equation

$$\sum y = an + b\sum x$$
$$\sum xy = a\sum x + b\sum x^2$$

derive the normal equations for the least square line.



The values of least square lines corresponding to $x_1, x_2 \ldots x_n$ are —

$$a + bx_1,\ a + bx_2 \ldots\ a + bx_n.$$

The corresponding vertical equations are

$$d_1 = a + bx_1 - y_1 \quad d_2 = a + bx_2 - y_2$$
$$\ldots\ d_n = a + bx_n - y_n.$$

Then the sum of the square of the deviations is —

$$d_1^2 + d_2^2 + \cdots + d_n^2 = (a+bx_1-y_1)^2 + (a+bx_2-y_2)^2 + \cdots$$
$$\cdots + (a+bx_n-y_n)^2$$

or $\sum d^2 = \sum (a+bx-y)^2$

R.H.S. is a fn. of a and b and can be written as —

$$F(a,b) = \sum (a+bx-y)^2$$

A necessary condition for this to be a minimum (or a maximum) is that

$$\frac{\partial F}{\partial a} = 0 \quad \text{and} \quad \frac{\partial F}{\partial b} = 0$$

$$\frac{\partial F}{\partial a} = \sum \frac{\partial}{\partial a}(a+bx-y)^2 = \sum 2(a+bx-y)$$

$$\frac{\partial F}{\partial b} = \sum \frac{\partial}{\partial b}(a+bx-y)^2 = \sum 2x(a+bx-y)$$

From the 1st Eqn. we get —

$$\sum (a+bx-y) = 0$$

i.e. $\sum a + \sum bx - \sum y = 0$

or $\sum y = an + b\sum x \quad \cdots\cdots (1)$

From the second equation we obtain

$$\sum x(a+bx-y) = 0 \Rightarrow \sum ax + \sum bx^2 - \sum xy = 0$$

$$\Rightarrow \sum xy = a\sum x + b\sum x^2 \quad \cdots\cdots (2)$$

Ex - 2

Prove that a least-square line always passes through the point $(\bar{x}, \bar{y})$.

~~Case I~~

x is the independent variable.

The equation of the least-square line is —

$$y = a + bx \qquad (1)$$

A normal equation for the least-square line is —

$$\sum y = an + b\sum x \qquad (2)$$

Dividing both sides of (2) by n gives —

$$\frac{\sum y}{n} = a + b\frac{\sum x}{n}$$

or $$\bar{y} = a + b\bar{x} \qquad -(3)$$

Subtracting (3) from (1) we get —

$$(y - \bar{y}) = b(x - \bar{x}) \qquad -(4)$$

which shows that the line passes through $\bar{x}$ and $\bar{y}$.

Ex - 3

~~Height x of Father (inches)~~	~~65~~	~~63~~	~~67~~	~~64~~	~~68~~	~~62~~
~~Height y of Son (inches)~~						

Height x of Father (inches)	Height y of Son (inches)
65	68
63	66
67	68
64	65
68	69
62	66
70	68
66	65
68	71
67	67
69	68
71	70

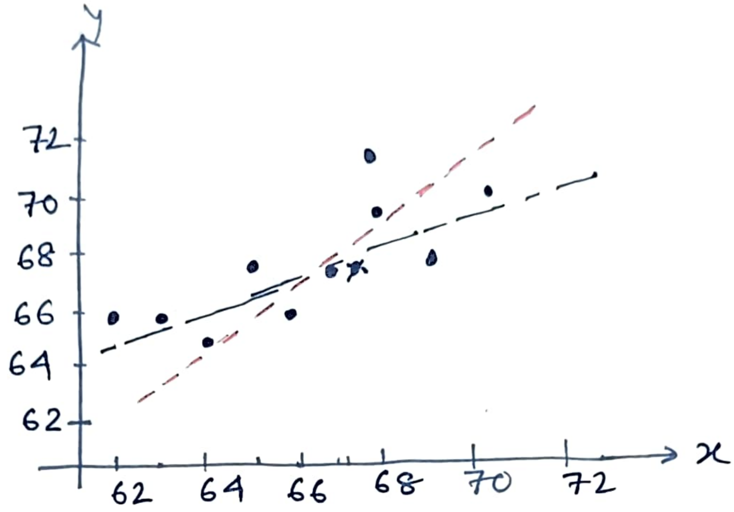
The table records the respective heights x and y of a sample of 12 fathers and their oldest sons.

a) Construct a scatter diagram.

b) Find the least-square ~~line~~ regression line of y on x.

c) Find the least-square regression line of x on y.

a)



b)

x	y	x^2	xy	y^2
65	68	4225	4420	4624
63	66	3969	4158	4356
67	68	4489	4556	4624
64	65	4096	4160	4225
68	69	4624	4692	4761
62	66	3844	4092	4356
70	68	4900	4760	4624
66	65	4356	4290	4225
68	71	4624	4828	5041
67	67	4489	4489	4489
69	68	4761	4692	4624
71	70	5041	4970	4900
$\sum x = 800$	$\sum y = 811$	$\sum x^2 = 53{,}418$	$\sum xy = 54{,}170$	$\sum y^2 = 54{,}849$

The regression line of y on x is giveen by-

$$y = a + bx$$

where a and b are obtained by solveing the normal equation-

$$\sum y = an + b\sum x$$

$$\sum xy = a\sum x + b\sum x^2$$

From the table. we obtain

$$811 = 12a + 800b \quad -(1)$$

$$800a + 53,418b = 54107 \quad -(2)$$

solveing the eqn. (1) and (2) we obtain-

$$a = 35.82 \quad \text{and} \quad b = 0.476$$

$\therefore$ the equation of the regression line is

$$y = 35.82 + 0.476x.$$

(c) The regression line of x on y is given by-

$$x = c + dy.$$

where c and d are obtained by solveing the normal equation

$$\sum x = cn + d\sum y$$

$$\sum xy = c\sum y + d\sum y^2$$

using the data given in the table we obtain

$$12c + 811d = 800$$

$$811c + 54,849d = 54,107.$$

solving this eqn. we get —

$$c = -3.38 \text{ and } d = 1.036.$$

$\therefore$ $x = -3.38 + 1.036y$ is the regression line of x on y.